

Internal Assessment 1 — Data Structures and Algorithms (CS201)

Time: 90 minutes

Maximum Marks: 32

Attempt all questions. Q4 and Q5 carry an internal choice: attempt any ONE.

SECTION A — Stacks and Queues

Q1.

Q1 (a) Define a stack and list two of its applications. **[3 marks]**

Q1 (b) Differentiate between a stack and a queue with suitable examples. **[4 marks]**

SECTION B — Searching and Graphs

Q2 Explain the binary search algorithm. State its precondition and derive its time complexity. **[8 marks]**

Q3.

Q3 (a) Write the algorithm for breadth first search (BFS) on a graph. **[4 marks]**

Q3 (b) Compare BFS and DFS with respect to the data structure used, time complexity and use cases. **[5 marks]**

SECTION C — Linked Lists and Hashing (Internal choice: answer Q4 OR Q5)

Q4 Explain how a singly linked list can be reversed. Give the algorithm and state its time and space complexity. **[8 marks]**

— OR —

Q5 Explain hashing. Describe any two collision resolution techniques. **[8 marks]**